

COMPARATIVE REVIEW OF NOVEL PHASE 1 ONCOLOGY TRIAL DESIGNS



A thesis submitted to the
Manipal Academy of Higher Education
For the degree of
Master of Science in Data Science (M.Sc.)

By,
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202410010

June 2022

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CERTIFICATE

This is to certify that Jagriti Bablani, a student of M.Sc. Data Science from Department of Data Science, PSPH, MAHE, Manipal has completed her fourth semester project under my guidance from January to June - 2022. The project work entitled “**Comparative Review of Novel Phase 1 Oncology Trial Designs**” embodies the novel work done.

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DECLARATION

I hereby declare that the project work entitled “Comparative Review of Novel Phase 1 Oncology Trial Designs” submitted to the Department of Data Science, PSPH, MAHE, Manipal, is a record of the work done by me under the guidance of Shreenidhi M (Associate Statistician) and Rupam Pal (Senior Manager - Statistics), Biostatistics - India, GSK,Bangalore.This thesis is submitted in partial fulfilment of the requirements for the award of degree of Master of Science in Data Science.

The results embodied in this thesis have not been submitted to any other Institute or University for the award of any degree/certificate.

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Acknowledgement

I take this opportunity to express my deep sense of thanks and gratitude to my project guides, Shreenidhi M and Rupam Pal, for providing their valuable guidance, comments, suggestions and constant encouragement throughout the project. Without their help, this work would never be a reality.

I owe a deep sense of gratitude to GlaxoSmithKline to provide me with the opportunity to do an internship with them. I would also like to extend my sincere thanks to Angshuman Sarkar, Head - Statistics, Biostatistics (India), GSK, Bangalore, for his support and guidance.

My heartfelt gratitude to Dr Asha Kamath, Head of the Department, Department of Data Science, for providing me with the opportunity and platform to present the project on the title mentioned above as a part of my curriculum. I thank all the faculty members, research scholars and office staffs of the Department of Data Science for their unconditional help throughout the course.

I am extremely thankful for the support and great love of my parents, cousins and friends, as they kept me going on, and this work wouldn't be possible without their solid moral support.

Finally, I am incredibly thankful to the Lord Almighty for his unconditional love and blessings. This accomplishment wouldn't be possible without the help of each one of them, who stood by me through thick and thin. Thank you.

Jagriti Bablani

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Abstract

The shifting landscape of oncology drug development requires a major change from the traditional clinical trial design paradigm, especially in the early stages. The major goal of a Phase I oncology clinical trial has always been to assess an investigational therapy's safety and tolerability by identifying the maximum tolerated dose (MTD). While this conventional Phase I clinical trial design approach is appropriate for testing cytotoxic agents, immune-oncology agents, such as immune checkpoint inhibitors, have different assumptions and challenges. As a result, trial designs and endpoints must be adapted to the particular drugs under investigation. Late-onset of toxicity is one of the major issues we confront when designing phase I trials for these immune-oncology therapies.

This thesis evaluates and assesses a new design like TITE-BOIN and compares them with some of the traditional methods like MTPI addressing late-onset toxicity using simulations, and then, if possible, give some recommendations on design choices under these various scenarios.

Chapter 1

Introduction

1.1 Clinical Trials

The research studies conducted on human subjects to evaluate medical or surgical procedures are called “clinical trials”. Clinical trials are being used by scientists to determine whether a new treatment is safe and effective in humans. A clinical trial is also conducted to determine whether another medication is more viable and has better safety and efficacy profile than the one under consideration. Scientists do laboratory experiments and examinations on animals to look at a possible drug’s well-being and adequacy before the U.S. administrative authority, the Food and Drug Administration (FDA), agrees to begin the clinical trial. The FDA grants permission to proceed with a human trial if the animal study shows promising results.

Typically, clinical trials progress through four stages namely Phase I, Phase II, Phase III, and Phase IV. In phase I, preliminary safety evaluation of a drug is carried out on a set number of healthy volunteers. The primary aim of phase I is to assess safety, tolerability of a drug and find the correct dose that possibly be taken forward into further phases of drug development.

The phase II trial has two sections: Phase IIA and Phase IIB. Phase IIA trials are generally pilot studies intended to show biological activity or clinical adequacy. Phase

IIB is specifically design to study efficacy to assess how the drug works at the prescribed dose(s). By drawing comparisons to standard medicine, the Phase III preliminaries establish the efficacy and safety of the treatment being commercialised. It involves a huge group of people, ranging in size from hundreds to thousands. The FDA requests Phase I, II, and III trial results to conclude whether the new drug can be given endorsement for the general usage in the patient population. In the wake of getting endorsement for the utilisation of medications or devices from the FDA, a phase IV trial will assess its safety and efficacy in large, diverse populations.

1.2 Oncology Clinical Trials

Cancer patients are used as subjects in Phase 1 studies of anti-cancer therapies, making the oncology study unique among clinical trials in other therapeutic areas. There are various kinds of anti-cancer agents: chemotherapeutic specialists (CAs), molecularly targeted agents (MTAs), immunotherapeutic agents (IAs), etc. Because of the high toxicity of anti-cancer agents, there is high risk of frequent adverse events. Patients with cancer benefit from their therapeutic effects, such as shrinking tumour size, increased general endurance, and so on. Disease patients with severe conditions for which prolonged life length can't be achieved or with side effects that can't be felt better utilising standard therapies are taken as subjects in the oncology trial, so they are in preparation to get the benefit of anti-cancer therapies disregarding the toxicity related to it to some extent. As a result, disease preliminaries are intended to decrease the probability of treating cancer patients with toxic doses or sub-therapeutic doses with a restricted sample size of cancer patients. If the reported adverse events or efficacy is low, the dose level should be increased, but promptly decreased if the observed toxicity reaction is severe or unreasonably

high. As a result, the goal is to treat disease patients enrolled in the experiment with the maximum efficacy and least toxicity possible. Consequently an anti-cancer drug's early-phase clinical trial is referred to as a "dose-finding" experiment because it determines the MTD, or the drug's which has maximum efficacy and acceptable toxicity. "Maximum tolerated dose (MTD) is the largest dose at which the patient can tolerate the resulting toxicities"[5]. There are novel designs which are developed to finding the MTD in Oncology field. In this project, two of those novel designs will be compared to evaluate their performance.

1.3 Objective

The objective is to do a comparative review of the **Modified Toxicity Probability Interval** (mTPI) and Time to Event **Bayesian Optimal INterval** design (TITE-BOIN) designs under various scenario addressing the **late onset of toxicity** problem.

Chapter 2

Methodology

2.1 Late onset of toxicity

Traditional dose-finding designs were at first created to evaluate Cytotoxic drugs (e.g., Chemotherapy). For chemotherapy drugs, the toxicities were frequently acute and can be effectively observed in the first cycle of therapy (28 days). So, traditional designs would be suitable for chemotherapy agents. However, the development of immunotherapies and molecularly targeted therapies have posed challenges since the toxicities could possibly occur after the first cycle of therapy. To account for the late onset of toxicity, long assessment windows must be used. Immuno-oncology and molecularly targeted therapies have assumptions and challenges that traditional designs cannot address. Traditional phase I trial designs that rely on early toxicity detection to apply dose-escalation criteria to new patients are complicated by this.

2.2 Modified toxicity probability interval

Yuan Ji introduced the MTPI methodology in 2010 as a modified version of the toxicity likelihood probability interval (TPI) method. It is a dose-finding design that utilizes the posterior inference for a straightforward Bayesian model and applies to dose-finding stud-

ies with a binary toxicity endpoint. This technique recommends a dose-finding decision rule in light of the unit probability mass (UPM). “For a given interval and probability distribution, the UPM of that interval is defined as the probability of the interval divided by the length of the interval”. For example, “for a continuous random variable X with cumulative distribution function $F(x)$, the UPM of an interval (a, b) under the distribution $F(x)$ is $\frac{F(b)-F(a)}{b-a}$ ”[8].

2.2.1 Dose-finding method

Let’s consider D dose levels of a particular medication. Let p_d be the unknown toxicity true probability associated with the d^{th} dose, $d = 1, \dots, D$. As the dose level increases probability of toxicity also increases $p_1 < p_2 \dots < p_D$. Let that dose d is presently utilized to treat patients and n_d be the patients at the dose ($n_d \geq 1$) out of which x_d ($x_d \leq n_d$) patients have the DLT. Let prior distribution for p_d is $\text{Beta}(\alpha_d, \beta_d)$. Using the observed values of x_d and n_d , posterior distribution for p_d is given by $\text{Beta}(\alpha_d + x_d, \beta_d + n_d - x_d)$ using the beta-binomial conjugacy. Based on the posterior, one of the following three choices needs to be made:

- de-escalate the dose to the next lower dose ($d-1$)
- retain the same dose d
- escalate the dose to the next higher dose ($d+1$)

Depending on the decision, the next cohort of patients are treated at dose $j \in d-1, d, d+1$.

Similarly, the process will go on until MTD is reached.

The “equivalence interval (EI) is defined as $[p_t - \epsilon_1, p_t + \epsilon_2]$, $\epsilon_1, \epsilon_2 \geq 0$ ”[6], where ϵ_1 and ϵ_2 are small fractions, such as 0.05, to account for the uncertainty around the true target

toxicity probability. “Target toxicity level is the maximum probability of toxicity that one is considered acceptable in the trial (p_t)”[2]. “ $p_t - \epsilon_1$ is the lowest toxicity probability that would be comfortable using to treat future patients without dose escalation”. “ $p_t + \epsilon_2$ is the highest toxicity probability that the one would be comfortable using to treat future patients without dose de-escalation”[6]. A dose, for which the true toxicity probability lies within this interval EI can be considered as potential true MTD. Different values of ϵ_1 and ϵ_2 can be used depending on the requirement of the disease under consideration and these values need to be pre-specified in the clinical trial protocol. The target p_t values are generally less than or equal to 30%. A value greater than 30% indicates that patients are being exposed to high-toxicity doses unnecessarily.

Determining an equality interval prompts partitioning the interval $(0, 1)$ into three sub-interval. The under-dosing interval is $(0, p_t - \epsilon_1)$, the proper dosing interval is $[p_t - \epsilon_1, p_t + \epsilon_2]$, and the over-dosing interval is $(p_t + \epsilon_2, 1)$.

This design also includes certain additional safety criteria that are necessary for practical issues. Let’s say, a cohort of patients have been administrated dose d_i , $i \in 1, \dots, D$ and dose limiting toxicities are being recorded. Then based on the observed data, we escalate to the following higher dose level if the interval $(0, p_t - \epsilon_1)$ has the biggest UPM, remain at that dose level if the interval $[p_t - \epsilon_1, p_t + \epsilon_2]$ has the biggest UPM, or de-escalate to next lower dose level if the interval $(p_t + \epsilon_2, 1)$ has the biggest UPM. Additionally 2 safety rule would be utilized.

- “Safety rule 1 (early termination) says that dose 1 has been used to treat patients. If $P(p_d \geq p_t \mid data) \geq \phi$ for a ϕ close to 1 (say, $\phi = 0.95$), then terminate the trial due to excessive toxicity”[6].
- “Safety rule 2 (dose exclusion) says that the decision is to escalate from dose d to

($d + 1$). If $P(p_{d+1} \geq p_T \mid data) \geq \phi$, for a ϕ close to 1 (say, $\phi = 0.95$), then treat the next cohort of patients at dose d and exclude doses ($d + 1$) and higher from the trial, that is, these doses will never be used again in the trial”[6].

At the end of the trial, for finding MTD we consider all the tried doses which satisfy the condition $P(p_d \geq p_T \mid data) \geq \phi$. Among these doses the one with smallest difference $|\hat{p}_d - p_T|$ is chosen as MTD. Here $\hat{p}_d$ is an estimate of p_d , which is calculated using the isotonically transformed posterior mean [6].

2.3 Time to event Bayesian optimal interval

TITE-BOIN (Time-to-event Bayesian optimal interval) design speeds up phase I trial by taking into account continuing dose assignment decisions for new patients while toxicity information for certain selected patients is still awaiting. The TITE-BOIN dose-escalation/de-escalation rule can be specified before the trial begins, making it simple and straightforward to perform. The TITE-BOIN has equivalent precision to the more complex model-based designs like the time-to-event continuous reassessment approach (TITE-CRM), but it is simpler to carry out with notably better overdose control[4]. "The TITE-BOIN automatically reduces to the BOIN design when there is no pending data"[4].

2.3.1 Dose-finding method

Let’s consider D dose levels of a particular medication. Let p_d be the unknown toxicity true probability associated with the d^{th} dose, $d = 1, \dots, D$. As the dose level increases probability of toxicity also increases $p_1 < p_2 \dots < p_D$. Let us assume that dose d is currently administrated to treat patients, n_d patients are treated in the cohort after each of these

n_d patients are followed up for at least pre-specified time (say 4 weeks) and at least 50% of patients have completed their assessment time in a particular dose level a decision will be made regarding the dose of the next cohort of the patients. This implies few of the n_d patients may not complete their full assessment time (say 8 weeks) before the decision is taken. Let r be the number of patients who completed the assessment, c be the patients who did not completed the assessment and S be the patients out of n_d who had DTL. “The total follow-up time (TFT) is defined as the sum of all currently pending patients’ follow-up times at the current dose, and standardised TFT (STFT) is defined as the TFT divided by the length of the DLT assessment window”[4].

Under the TITE-BOIN design, the dose escalation and de-escalation decision are determined by comparing the estimate true toxicity probability of a particular dose d , p_d given by

$$\hat{p}_d = \frac{S + \frac{p_0}{1+p_0}(c - STFT)}{n} \quad (2.1)$$

with the dose escalation and de-escalation boundaries.

Equation 2.1 has a unknown value p_0 which would be estimated using Bayesian posterior mean $\tilde{p}_0$ in light of the observed information. “Assuming the beta-binomial model with the prior distribution $p_0 \sim \text{Beta}(\alpha, \beta)$ the bayesian posterior mean estimate is given by $\tilde{p}_0 = (s + \alpha) / (r + \alpha + \beta)$ ”[4], with the dose escalations and escalation decisions as define as below.

- de-escalate the dose to the next lower dose ($d-1$), if $\hat{p}_d \geq \lambda_d$
- retain the same dose d , if $\lambda_d < \hat{p}_d < \lambda_d$
- escalate the dose to the next higher dose ($d+1$), if $\hat{p}_d \leq \lambda_e$

“Given the target DLT rate of ϕ and assuming the noninformative prior the current dose is equally likely to be below, equal to, or above the MTD, the optimal escalation and de-escalation boundaries are found and λ_e and λ_d that minimize the incorrect decision of dose escalation and de-escalation are given by”[4]

$$\lambda_e = \frac{\log \frac{1-\phi_1}{1-\phi}}{\log \frac{\phi(1-\phi_1)}{\phi_1(1-\phi)}} \quad (2.2)$$

$$\lambda_d = \frac{\log \frac{1-\phi}{1-\phi_2}}{\log \frac{\phi_2(1-\phi)}{\phi(1-\phi_2)}} \quad (2.3)$$

“where ϕ is the highest toxicity probability that is deemed subtherapeutic(i.e., below the MTD)”[4] to such an extent that dose escalation ought to be made, and ϕ_2 is the lowest toxicity probability that is considered excessively toxic such that dose de-escalation is required. “Liu and Yuan (2016) recommend default values $\phi_1 = 0.6\phi$ and $\phi_2=1.4\phi$ for general use”[4].

The TITE-BOIN design incorporates data from all patients, at the current dose d including DLT data from patients who have finished entire assessment time (say 8 weeks) and follow-up time data from patients who still are completing their assessment time. “During trial conduct, the following overdose control safety stopping rule is imposed: if $\Pr(p < \phi \mid s, n) < 0.95$ and $n \leq 3$, eliminate the current and higher doses from the trial; if the lowest dose is eliminated, terminate the trial early for safety”[4].

Chapter 3

Simulation

1000 trials of each design are simulated under different scenarios using the R language, and operating characteristics are obtained to compare MTPI with TITE-BOIN. “Number of patients treated at a dose level is known as cohort size and is chosen to be 3 in the simulation”[2]. The maximum sample size for a trial is kept at 50. “Starting dose is the dose chosen to treat the first cohort of patients in a phase I trial”[2]. The trial starts from the very beginning, with first dose. The toxicity response is then generated using True probability of toxicity with an 8-week DLT window. Once the data is generated, the dose escalation and escalation rules will be utilized to find the next dose level. Same procedure would be repeated until the termination rules are satisfied. Detailed algorithm are discussed in the Algorithm part A.

For MTPI design, the prior distribution for p_i considered for simulation is Beta(1, 1). The target toxicity rate, p_T is taken as 0.28 and $\epsilon_1 = \epsilon_2 = 0.05$. The under-dosing interval considered is (0, 0.23). Proper dosing interval considered is [0.23, 0.33]. Over-dosing interval (0.33,1).

In MTPI design 2 scenario's are considered. First is MTPI (8 weeks) that is the follow up time is 8 weeks for each patient and DLT window of 8 weeks. Second MTPI (4 weeks) that is the follow up time is 4 weeks for each patient with is generally used in MTPI designs and DLT window is 8 weeks. Which is DLT is happing after 4 weeks it will

not be considered as DLT since subjects are followed up for assessment until 4 weeks.

Algorithm A.1 contains the MTPI design algorithm.

For TiTE-BOIN ϕ is considered as 0.28 which gives decision boundaries λ_e and λ_d as 0.2364907 and 0.3585195 respectively, the prior distribution for p_i considered for simulation is Beta(0.15,0.85). Assessment time is 8 weeks for each patient and DLT window of 8 weeks and each patient in particular cohort should be enrolled and should complete at least 4 weeks of follow up time before decision making. Algorithm A.2 contains the TITE-BOIN design algorithm.

3.1 Scenarios

3.1.1 Toxicity Scenarios

The number of dose levels selected for simulations is five under which different scenarios are taken for simulation such as:

Toxicity Scenario	P1	P2	P3	P4	P5
5 dose levels					
All Doses are safe	0.01	0.05	0.1	0.2	0.25
Low Toxicity	0.01	0.05	0.1	0.28	0.4
Moderate Toxicity	0.05	0.1	0.28	0.4	0.45
High Toxicity	0.1	0.28	0.4	0.45	0.5
All Doses are Toxic	0.35	0.40	0.45	0.50	0.55

All doses are safe means all the doses are lower than true MTD ,in Low toxicity scenarios the actual MTD is considered to be at higher dose levels, Moderate toxicity scenarios considered to be at the moderate dose, High toxicity actual MTD is considered to be at lower dose levels, and All doses are toxic means all the doses are higher than true MTD.

3.1.2 Enrollment Scenarios

Different enrollment scenarios for simulation are considered such as:

Enrollment Scenario	Enrollment Time
Rapid	All the patients will be enrolled within 3 weeks.
Moderate	All the patients will be enrolled within 8 weeks.
Slow	All the patients will be enrolled within 12 weeks.

In Rapid enrollment all the patients in the cohort will be enrolled quickly, Moderate enrollment patients will be moderate time, and in slow enrollment patients in the cohort will be enrolled very slowly.

3.2 Operating characteristics

“The operating characteristics (OCs) considered for the comparison of candidate designs are the following”[7].

- Percentage of trials selecting as MTD
 - Dose below MTD
 - MTD
 - Dose beyond MTD
- Percentage of trials all doses are safe
- Percentage of trials all doses are toxic
- Average time taken
- Average sample of size
- Average Percentage of patients overdosed

Chapter 4

Results

4.1 Performance in selection of MTD for 5 doses Rapid Enrollment

Dose	Low Toxicity			Moderate Toxicity			High Toxicity		
	Trials Selecting Dose as MTD (%)			Trials Selecting Dose as MTD (%)			Trials Selecting Dose as MTD (%)		
	MTPI 8 Weeks	TITE BOIN	MTPI 4 Weeks	MTPI 8 Weeks	TITE BOIN	MTPI 4 Weeks	MTPI 8 Weeks	TITE BOIN	MTPI 4 Weeks
Dose below MTD	28.1	22.7	6.9	0.43	22.1	5.3	21	9.8	5.5
MTD	51.8	49.6	26.6	52.4	44.8	25.6	51.3	45.8	26.5
Dose beyond MTD	16.7	21.6	35.8	20.1	25.1	52.4	23.9	28.3	58.7
Trials Selecting All Doses as Safe (%)	3.2	5.1	30.4	0.4	0.8	16.5	0	0	8.3
Trials Selecting All Doses as Toxic (%)	0	1	0	0.7	7.2	0.2	3.8	16.1	1
Average time taken (In weeks)	71.25	61.72	46.11	61.26	50.5	43.48	50.49	39.56	39.04
Average Sample size	21.138	23.616	22.083	18.24	19.767	20.799	15.099	15.885	18.684
Average Overdosed patients (%)	11.77	13.14	27.69	15.84	17.95	39.51	22.25	24.15	49.29

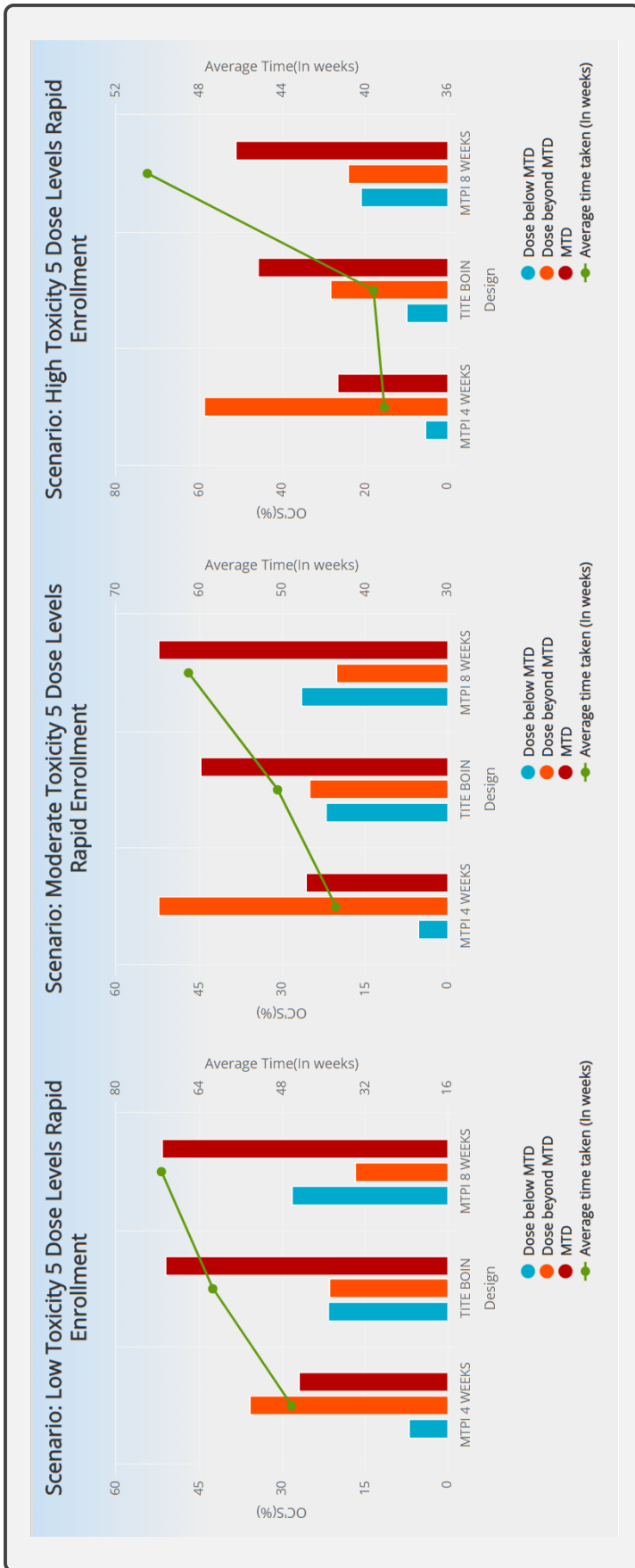


Figure 4.1: Performance in selection of MTD for 5 doses Rapid Enrollment

4.2 Performance in selection of MTD for 5 doses Moderate Enrollment

True DLT Rate	Low Toxicity			Moderate Toxicity			High Toxicity		
	Trials Selecting Dose as MTD (%)			Trials Selecting Dose as MTD (%)			Trials Selecting Dose as MTD (%)		
	MTPI 8 Weeks	TITE BOIN	MTPI 4 Weeks	MTPI 8 Weeks	TITE BOIN	MTPI 4 Weeks	MTPI 8 Weeks	TITE BOIN	MTPI 4 Weeks
Dose below MTD	28.3	18.3	6.9	26.4	16.3	5.3	21	6.4	5.5
MTD	51.8	49.7	26.9	52.4	45.2	25.6	51.3	44.3	26.5
Dose beyond MTD	16.7	25.4	35.8	20.1	26.1	52.4	23.9	29	58.7
Trials Selecting All Doses as Safe (%)	3.2	5.2	30.4	0.4	0.6	16.5	0	0.2	8.3
Trials Selecting All Doses as Toxic (%)	0	1.4	0	0.7	11.8	0.2	3.8	20.1	1
Average time taken (In weeks)	117.39	113.39	95.03	100.85	92.29	89.33	82.94	75.89	80.02
Average Sample size	21.138	24.855	22.083	18.24	20.355	20.799	15.099	16.788	18.684
Average Overdosed patients (%)	11.76	15	27.68	15.8	20.5	39.5	22.25	29.02	49.29

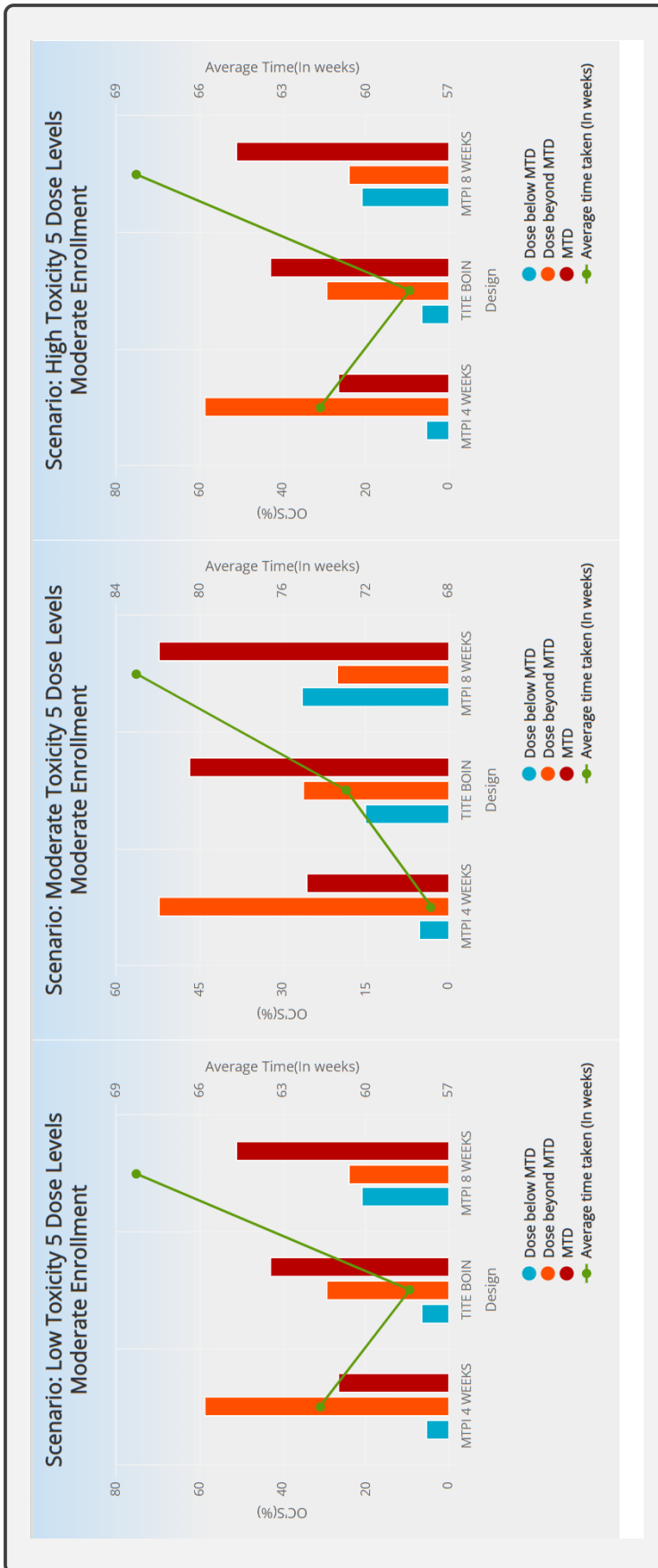


Figure 4.2: Performance in selection of MTD for 5 doses Moderate Enrollment

4.3 Performance in selection of MTD for 5 doses Slow Enrollment

Dose	Low Toxicity				Moderate Toxicity				High Toxicity			
	Trials Selecting Dose as MTD (%)				Trials Selecting Dose as MTD (%)				Trials Selecting Dose as MTD (%)			
	MTPI 8 Weeks	TITE-BOIN	MTPI 4 Weeks	MTPI 8 Weeks	MTPI 8 Weeks	TITE-BOIN	MTPI 4 Weeks	MTPI 4 Weeks	MTPI 8 Weeks	TITE-BOIN	MTPI 4 Weeks	MTPI 4 Weeks
Dose below MTD	28.3	18.3	6.9	26.4	26.4	16.3	5.3	5.3	21	6.4	5.5	5.5
MTD	51.8	49.7	26.9	52.4	52.4	45.2	25.6	25.6	51.3	44.3	26.5	26.5
Dose beyond MTD	16.7	25.4	35.8	20.1	20.1	26.1	52.4	52.4	23.9	29	58.7	58.7
Trials Selecting All Doses as Safe (%)	3.2	5.2	30.4	0.4	0.4	0.6	16.5	16.5	0	0.2	8.3	8.3
Trials Selecting All Doses as Toxic (%)	0	1.4	0	0.7	0.7	11.8	0.2	0.2	3.8	20.1	1	1
Average time taken (In weeks)	117.39	113.39	95.03	100.85	100.85	92.29	89.33	89.33	82.94	75.89	80.02	80.02
Average Sample size	21.138	24.855	22.083	18.24	18.24	20.355	20.799	20.799	15.099	16.788	18.684	18.684
Average Overdosed patients (%)	11.76	15.09	27.68	15.83	15.83	20.57	39.50	39.50	22.25	29.02	49.29	49.29

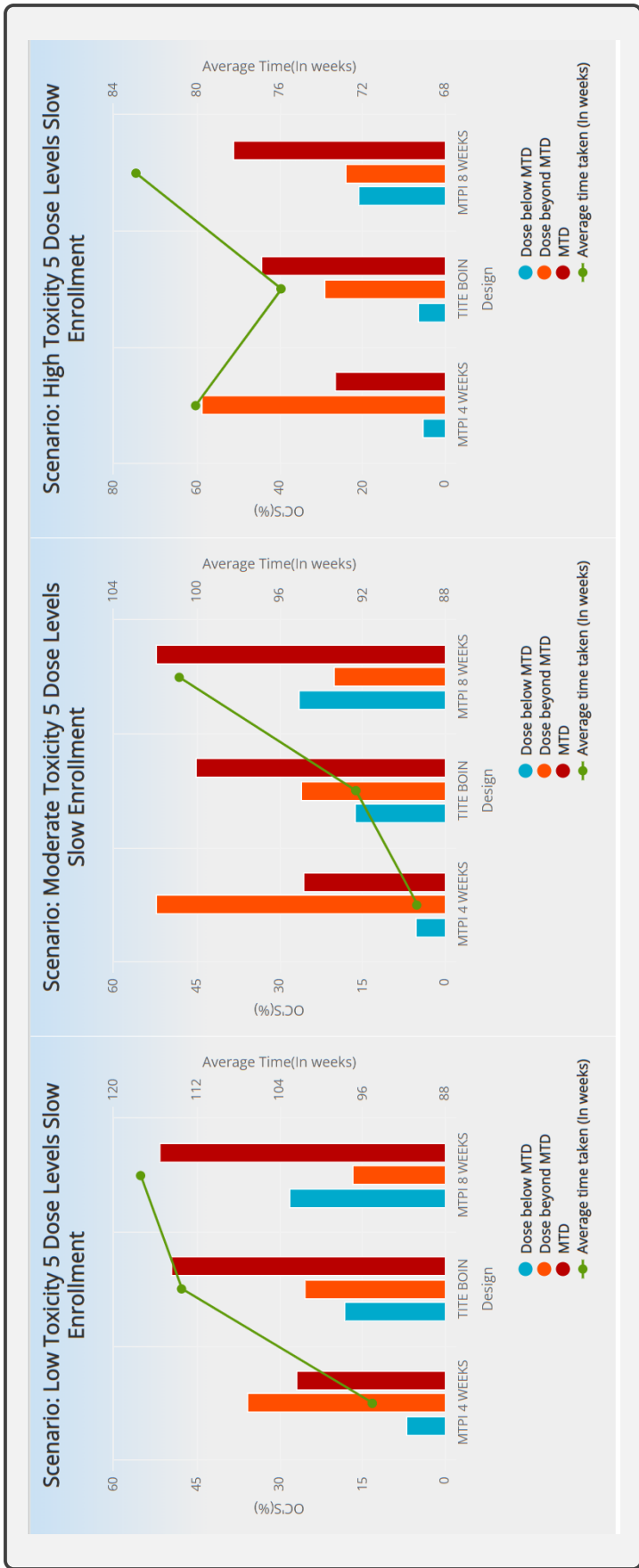


Figure 4.3: Performance in selection of MTD for 5 doses Slow Enrollment

4.4 Performance in selection of MTD for 5 doses when all doses are safe

True DLT Rate	Rapid Enrollment				Moderate Enrollment				Slow Enrollment			
	Trials Selecting Dose as MTD (%)				Trials Selecting Dose as MTD (%)				Trials Selecting Dose as MTD (%)			
	MTPI 8 Weeks	TITE-BOIN	MTPI 4 Weeks	MTPI 8 Weeks	MTPI 8 Weeks	TITE-BOIN	MTPI 4 Weeks	MTPI 8 Weeks	MTPI 8 Weeks	TITE-BOIN	MTPI 4 Weeks	MTPI 8 Weeks
0.01	0.4	0	0.2	0.4	0.4	0	0.2	0.4	0.4	0	0.2	0.2
0.05	5.4	3.9	1.8	5.4	5.4	2.6	1.8	5.4	5.4	2.6	1.8	1.8
0.1	16.9	15.1	4.9	16.9	16.9	12.6	4.9	16.9	16.9	12.6	4.9	4.9
0.2	34.7	32	14	34.7	34.7	29.8	14	34.7	34.7	29.8	14	14
0.25	27.7	28.7	22	27.7	27.7	32.7	22	27.7	27.7	32.7	22	22
Average Sample size	21.393	23.067	22.248	21.393	21.393	24.027	22.248	21.393	21.393	24.027	22.248	22.248
Average time taken (In weeks)	72.4	61.22	46.26	98.31	98.31	88.46	73.63	119.3	119.3	88.46	95.71	95.71
Trials Selecting All Doses as Safe (%)	14.9	18.8	57.1	14.9	14.9	19.7	57.1	14.9	14.9	19.7	57.1	57.1
Trials Selecting All Doses as Toxic (%)	0	1.5	0	0	0	2.6	0	0	0	2.6	0	0

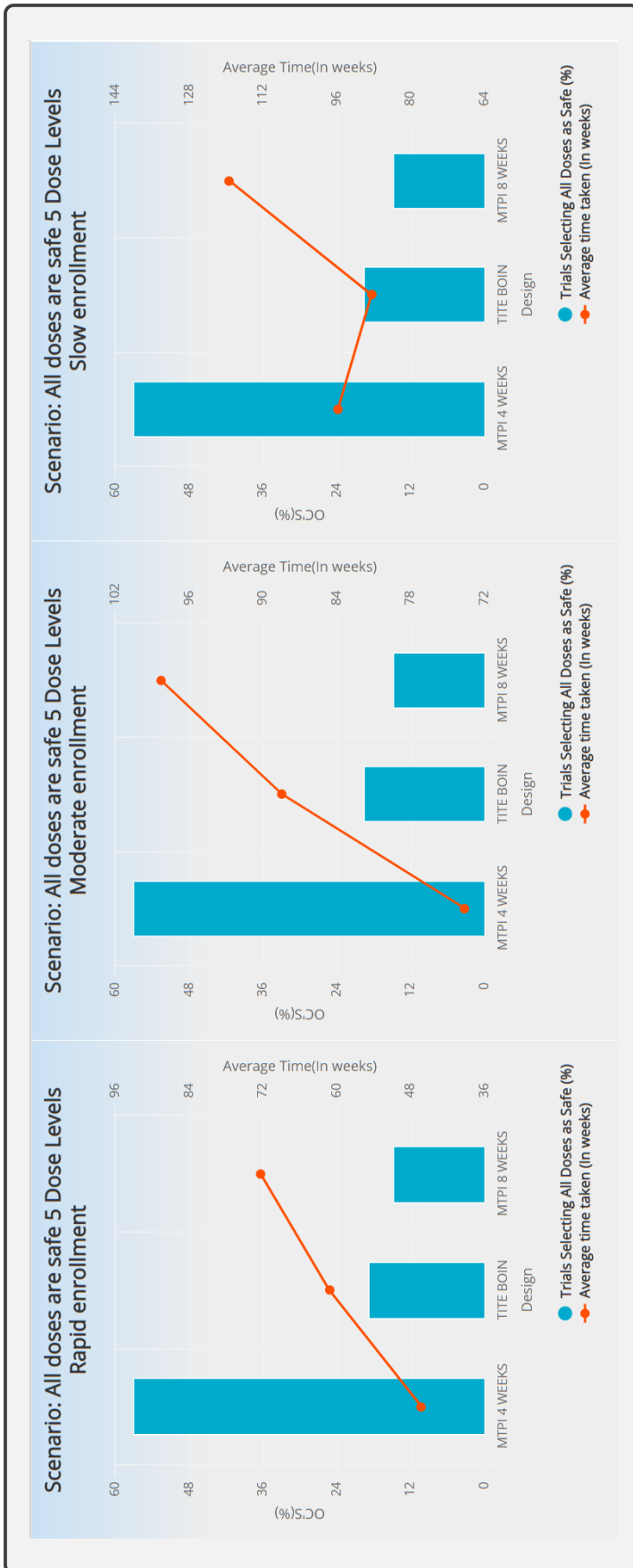


Figure 4.4: Performance in selection of MTD for 5 doses when all doses are safe

4.5 Performance in selection of MTD for 5 doses when all doses are toxic

True DLT Rate	Rapid Enrollment				Modrate Enrollment				Slow Enrollment			
	Trials Selecting Dose as MTD (%)				Trials Selecting Dose as MTD (%)				Trials Selecting Dose as MTD (%)			
	MTPI 8 Weeks	TITE-BOIN	MTPI 4 Weeks	MTPI 8 Weeks	MTPI 8 Weeks	TITE-BOIN	MTPI 4 Weeks	MTPI 8 Weeks	MTPI 8 Weeks	TITE-BOIN	MTPI 4 Weeks	MTPI 8 Weeks
0.35	39.3	10.3	25.5	39.3	39.3	7.2	25.5	39.3	39.3	7.2	25.5	25.5
0.4	12.3	14.5	24.1	12.3	12.3	13.6	24.1	12.3	12.3	13.6	24.1	24.1
0.45	3.3	3.2	16.5	3.3	3.3	5	16.5	3.3	3.3	5	16.5	16.5
0.5	0.4	0.5	11.5	0.4	0.4	0.5	11.5	0.4	0.4	0.5	11.5	11.5
0.55	0	0.2	7.4	0	0	0.2	7.4	0	0	0.2	7.4	7.4
Average Sample size	8.502	8.067	14.673	8.502	8.502	7.986	14.673	8.502	8.502	7.986	14.673	14.673
Average time taken (In weeks)	27.86	20.04	30.77	37.54	37.54	35.98	48.34	45.62	45.62	35.98	62.7	62.7
Trials Selecting All Doses as Safe (%)	0	0	3.1	0	0	0	3.1	0	0	0	3.1	3.1
Trials Selecting All Doses as Toxic (%)	44.7	71.3	11.9	44.7	44.7	73.5	11.9	44.7	44.7	73.5	11.9	11.9

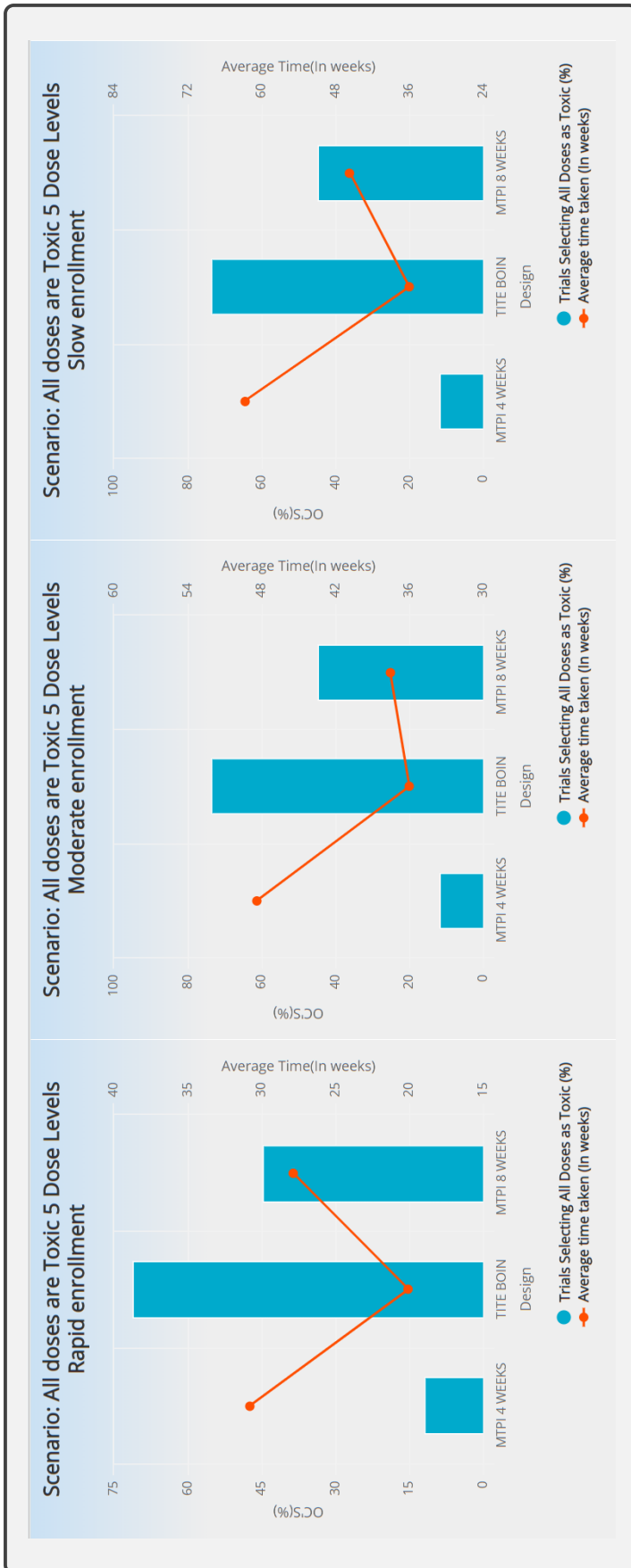


Figure 4.5: Performance in selection of MTD for 5 doses when all doses are toxic

Chapter 5

Comparison

In the case of low toxicity and rapid enrollment, MTPI 4 weeks (Assessment time of 4 weeks) chooses the proper dose (0.28) as MTD 26.9% and toxic doses 35.5% of the trials, MTPI 8 weeks (Assessment time of 8 weeks) is 51.8% and 16.7%, and TITE-BOIN is 49.6% and 21.5, with an average time to complete the trial of 46.11, 71.2, and 61.72 weeks respectively. MTPI (4 weeks) takes less time to complete the trial chooses toxic doses as MTD and increases risk of patients getting exposed to toxic doses. MTPI (8 weeks) and TITE-BOIN there both chooses proper dose more often but TITE-BOIN takes less time when compared to MTPI (8 weeks). In all the scenario's of rapid enrollment and moderate enrollment we can see that MTPI (4 weeks) is completing the trial early but selecting toxic dose as MTD more percentage of time. In MTPI (8 weeks) and TITE-BOIN there is no significant difference between selection of correct dosage as MTD. In addition TITE-BOIN is completing the trial early when compared to MTPI (8 weeks).

When the enrollment is slow and toxicity is high, MTPI(4 weeks) chooses the right dosage (0.28) as MTD 26.5% and toxic dosage 58.5% of the trials, MTPI (8 weeks) is 51.3% and 23.9%, and TITE-BOIN is 44.3% and 29%, with an average time to complete the trial of 80.2, 82.94, and 75.89 weeks respectively. There is no significant difference between the completion of MTPI (4 weeks), MTPI (8 weeks), and TITE-BOIN but MTPI (8 weeks) is performing better than MTPI (4 weeks) and TITE-BOIN it's choosing the right dose

(0.28) as MTD and reducing risk exposing patients to the toxic dose.

Let's consider the case of when all doses are safe and doses are toxic MTPI (4 Weeks) is given selecting all doses are safe 51% of the time MTPI (8 Weeks) is 14.9% and TITE-BOIN is 18.8%. MTPI (4 Weeks) is chosen correctly is significantly higher percentages of times when compared to MTPI (8 weeks) and TITE-BOIN but these results can be misleading because in the case of when all doses are toxic MTPI (4 weeks) fails to choose correctly. In both cases, TITE-BOIN is giving better results with less time.

Overall, the TITE-BOIN design has comparable characteristics with MTPI 8 weeks with a range of 7-10 weeks less time in rapid and moderate enrolment scenarios. Whereas in the slow enrolment scenarios, the average time taken to complete the trials is more or less same for both TITE BOIN and MTPI (8 weeks). In addition, the percentage of time the toxic doses are administered is little more in TITE-BOIN compared to MTPI (8 weeks) which needs to be carefully studied while designing a study.

Chapter 6

Conclusion

Based on the simulation results we can say that TITE-BOIN is very close to MTPI (8 weeks) in selecting the correct dose as MTD. Moreover, the trial that uses the TITE-BOIN design takes less time to complete compared to those who use MTPI (8 weeks). Thus TITE-BOIN design address the problem of late-onset toxicity by producing fairly comparable results as produced by MTPI (8 weeks) method and in the process taking significantly less time to complete the trials. When compared to MTPI (4 weeks) TITE-BOIN always performs better as it exposes fewer patients to toxic doses and identifies the correct dose as MTD more frequently. The MTPI (4 weeks), although takes lesser time to complete, completely fails to address the problem of late-onset of toxicity in almost all the scenarios.

In the case of low and moderate toxicity and rapid and moderate enrollment, TITE-BOIN is showing similar results as MTPI (8 weeks) and its taking less time as well. But in case of high toxicity and slow enrollment TITE-BOIN results are not as good as MTPI (8 weeks) and there is no significant difference in the completion time of both the designs. When all the doses are toxic, TITE-BOIN outperforms MTPI (8 weeks) and MTPI (4 weeks) in giving the right decision while taking less time.

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Appendix A

Algorithm

A.1 MTPI

The total number of doses is D , and N is the number of simulated trials. The maximum sample size for a study is N_{max} , and the maximum number of patients at a given dose level is n_{max} . Over-dosing, Proper-dosing, and under-dosing intervals are represented by UPM1, UPM2, and UPM3, respectively.

- Step 1: Simulate toxicity data for n patients at the current dose level d_i using toxicity probability p_i . Let x = no. of patients with toxicity out of these n patients simulated.
- Step 2: Calculating the time if enrollment of each patient. From Uniform distribution (e.g 0,4)
- Step 3: For x (no. of patients with toxicity out of these three patients simulated) checking which patient had DLT and what is the Time To DLT.
 - Time to DLT is randomly sampled
 - For patient with DLT Time To DLT Is added for that patient and Total assessment time is Enrollment Time + Time To DLT

- For patient without DLT. DLT window Time is added (8 weeks) Is added for that patient and Total assessment time is Enrollment Time + DLT window Time
- Step 4: Maximum of Total assessment time is taken as Total time for that cohort.
- Step 5: Calculate p_d^s posterior distribution.
- Step 6: Check if $(i + 1)^{th}$ dose is unacceptably toxic: $\Pr (p_i \geq p_T \mid \text{data}) < 0.95$; if it is true, stay at i^{th} dose level and exclude $(i+1)$ and higher dose levels from the trial.
 - If at the first dose then no dose is selected as MTD all dose are toxic.
- Step 7: Find the UPM for three intervals.
 - UPM1 – Over-dosing Interval
 - UPM2 – Proper-dosing- Interval
 - UPM3 – Under-dosing

Set UPM3=0 to ensure that no dose escalation happens if the criteria in step 6 are met.

 - If $d = 1$ and $\Pr (p_i \geq p_T \mid \text{data}) < 0.95$, stop the experiment if no MTD dose is chosen, and claim "all doses are toxic" otherwise.
 - (i) If $UPM1 > UPM1$ $UPM1 > UPM2$ then, stop the trial, stop the experiment if no MTD dose is chosen, and claim "all doses are toxic".
 - (ii) If $UPM2 > UPM1$ & $UPM2 > UPM3$ & n_{max} at d is not reached $\rightarrow d^{th}$ dose.
 - (iii) If $UPM3 > UPM1$ & $UPM3 > UPM2 \rightarrow (d + 1)^{th}$ dose.

- If $d = D$, then
 - (i) If $UPM1 > UPM2$ & $UPM1 > UPM3 \rightarrow (d - 1)^{th}$ dose.
 - (ii) If $UPM2 > UPM1$ & $UPM2 > UPM3$ & n_{max} at d is not reached $\rightarrow$
 d^{th} dose
 - (iii) If $UPM3 > UPM1$ & $UPM3 > UPM2 \rightarrow (d + 1)^{th}$ dose
- Step 8: Stop simulating if following conditions hold's true
 - N_{max} is reached
 - $1 \leq d \leq D$, decision is to stay and n_{max} is reached
- Step 9: After stopping conditions are met select MTD.
- Step 10: Repeat Step 1 to Step 9 N number's of times.
- Step 11: Derive operating characteristics for MTPI Design using the simulated data.
- Step 12: Follow same procedure for all the scenario's

A.2 TITE-BOIN

The total number of doses is D , and N is the number of simulated trials. The maximum sample size for a study is N_{max} , and the maximum number of patients at a given dose level is n_{max} .

- Step 1: Simulate toxicity data for n patients at the current dose level d_i using toxicity probability p_i . Let x = no. of patients with toxicity out of these n patients simulated.

- Step 2: Calculating the time if enrollment of each patient. From Uniform distribution.
- Step 3: For x (no. of patients with toxicity out of these three patients simulated) checking which patient had DLT and what is the Time To DLT.
 - Time to DLT randomly sampled.
 - For patient with DLT Time To DLT Is added for that patient and Total assessment time is Enrollment Time + Time To DLT.
 - For patient without DLT. DLT window Time is added (8 weeks) Is added for that patient and Total assessment time is Enrollment Time + DLT window Time.
- Step 4: Sort the following data w.r.t assessment time and store it one variable (This variable will store all the data from all the cohorts).
- Step 5: Collect all the data for current dose level d_i and create a copy of it.
- Step 6: Check at what time 50% or more then 50% of the data is completed for a current dose. Using that calculated TFT
 - TFT= Time of 50% of completion of data- (enrollment time of pending patients)
 - If TFT is less, then 4 (that is each patient should be followed up at least for 4 weeks before making decision at that particular dose) wait for some time till all the enrolled patients complete 4 weeks of assessment time.
 - Add that extra time in total assessment time.

- Use the Copy data how many others are completing their assessment at the time of making decision.
- If all of them are completing, then TITE-BOIN converts to BOIN check for those conditions.
- Step 6: Break the data into 2 parts patients who have completed their assessment at the time of decision making and patients who are still pending.
- Step 7: Check if $\Pr(p < \phi \mid s, n) < 0.95$ and $n \leq 3$, eliminate the current and higher doses from the trial; if the lowest dose is eliminated, terminate the trial early for safety
- Step 8: Calculate $\hat{p}_d$
 - de-escalate to the next lower dose (d-1), if $\hat{p}_d \geq \lambda_d$
 - stay at the same dose d, if $\lambda_d < \hat{p}_d < \lambda_{d+1}$
 - escalate to the next higher dose (d+1), if $\hat{p}_d \leq \lambda_e$
- Step 9: Stop simulating if following conditions hold's true
 - Nmax is reached
 - $1 \leq d \leq D$, decision is to stay and nmax is reached
- Step 10: After stopping conditions are met select MTD.
- Step 11: Repeat Step 1 to Step 10 N number's of times.
- Step 12: Derive operating characteristics for MTPI Design using the simulated data.
- Step 13: Follow same procedure for all the scenario's

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